1.5 Discrete Random Variables

Discrete random variable

*The **discrete random variable** is defined as*

$$X(\Omega) = \{x_n\}_{n \in I}$$

where $I = \{1, 2, \dots\}$ is an ordered countable set.

Probability mass function of discrete R.V.

*The **probability mass function (PMF)** of a discrete random variable is defined as*

$$P_1 = P_X(x_1) = \mathbb{P}(X = x_1) = \sum_{\omega : X(\omega) = x_1} \mathbb{P}(\omega),$$
$$P_2 = P_X(x_2) = \mathbb{P}(X = x_2) = \sum_{\omega : X(\omega) = x_2} \mathbb{P}(\omega),$$
$$\vdots$$

and in general, we have

$$P_n = P_X(x_n) = \mathbb{P}(X = x_n) = \sum_{\omega : X(\omega) = x_n} \mathbb{P}(\omega).$$

In compact notation, we write $P_X : \mathbb{R} \to R_X = [0, 1]$ where

$$P_X(x) = \begin{cases} \mathbb{P}(X = x), & \text{if } x \in R_X \\ 0, & \text{if } x \notin R_X \end{cases},$$

where R_X is called support of the random variable X, that is, all possible values that X can take.

Cumulative distribution function of discrete R.V.

*The **cumulative distribution function (CDF)** of a discrete random variable is defined as*

$$F_X(x_n) = \sum_{m \in I : x_m \le x_n} P_X(x_m) = \sum_{m \in I : x_m \le x_n} \sum_{\omega : X(\omega) = x_m} \mathbb{P}(\omega).$$

and F_X is a nondecreasing and right continuous step function. Note that the above definition is an alternative way to write

$$F_X(x_n) = \mathbb{P}(X \le x_n),$$

where we defined earlier for random variables.

Expectation of discrete R.V.

*The **expectation** of a discrete random variable with respect to a probability measure $\mathbb{P}$ is defined as*

$$E(X) = \sum_{\omega\in\Omega} X(\omega)\mathbb{P}(\omega) = \sum_{n\in I} x_n \sum_{\omega:X(\omega)=x_n} \mathbb{P}(\omega),$$

or

$$E(X) = \sum_{n\in I} x_n \mathbb{P}(X = x_n) = \sum_{n\in I} x_n P_X(x_n) = \sum_{n\in I} x_n P_n.$$

For simplicity, we write

$$E(X) = \sum_x x P_X(x).$$

Properties of $E(X)$

Let $E(X)$ be the expectation of a discrete random variable $X : \Omega \to \mathbb{R}$ and let discrete random variable $Y = g(X)$ where $g : \mathbb{R} \to \mathbb{R}$. Then

1. $E(Y) = \sum_y y P_Y(y) = \sum_x g(x) P_X(x)$,
2. $E(aX) = aE(X), a \in \mathbb{R}$,
3. $E(aX + b) = aE(X) + b, a, b \in \mathbb{R}$.

Variance of discrete R.V.

*The **variance** of a discrete random variable is defined as*

$$\begin{aligned}
Var(X) &= \sum_{n\in I} (x_n - E(X))^2 P_n \\
&= \sum_{n\in I} \left(x_n^2 - 2x_n E(X) + (E(X))^2\right) P_n \\
&= \sum_{n\in I} x_n^2 P_n - \sum_{n\in I} 2x_n E(X) P_n + \sum_{n\in I} (E(X))^2 P_n \\
&= E\left(X^2\right) - 2E(X) \sum_{n\in I} x_n P_n + (E(X))^2 \sum_{n\in I} P_n \\
&= E\left(X^2\right) - 2\left(E(X)\right)^2 + \left(E(X)\right)^2 \\
&= E\left(X^2\right) - \left(E(X)\right)^2.
\end{aligned}$$

Properties of $Var(X)$

Show that

1. $Var(X) \geq 0$,
2. $Var(aX + b) = a^2 Var(X)$,

Standard deviation

*The **standard deviation** of a discrete random variable X is defined as*

$$\sigma_X = \sqrt{Var(X)}.$$

Example 1.6 Consider an experiment of a coin toss. Define a discrete random variable $X : \Omega \to \mathbb{R}$ where $\Omega = \{H, T\}$ and $\mathcal{F} = \{\Omega, \emptyset, \{H\}, \{T\}\}$ such that

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = T \\ 1, & \text{if } \omega = H \end{cases}.$$

Also, define a probability measure

$$\mathbb{P}(\Omega) = 1, \quad \mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}.$$

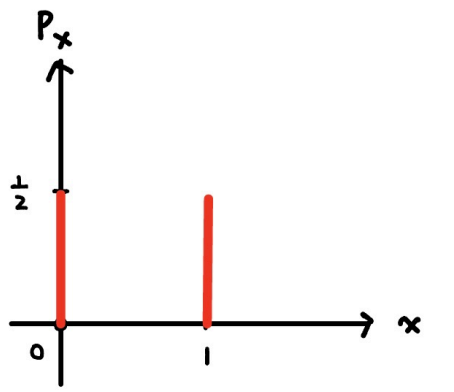
Then we have

1. PMF

$$P_1 = P_X(x_1) = \mathbb{P}(X = 0) = \mathbb{P}(\{T\}) = \frac{1}{2},$$
$$P_2 = P_X(x_2) = \mathbb{P}(X = 1) = \mathbb{P}(\{H\}) = \frac{1}{2}.$$

   Thus, we get

$$P_X(x) = \begin{cases} \frac{1}{2}, & \text{if } x = 0 \\ \\ \frac{1}{2}, & \text{if } x = 1 \end{cases}.$$

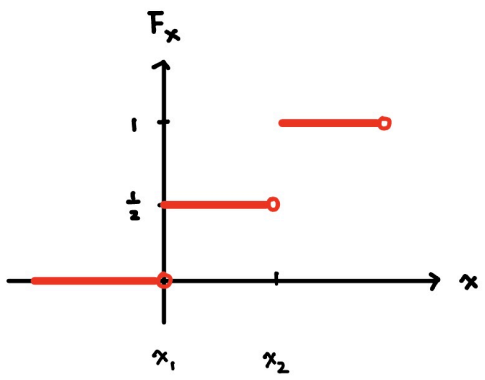


2. CDF

$$F_X(x_1) = \mathbb{P}(X \leq x_1 = 0) = P_1 = \frac{1}{2},$$
$$F_X(x_2) = \mathbb{P}(X \leq x_2 = 1) = P_1 + P_2 = 1.$$

   Thus, we get

$$F_X(x) = \begin{cases} \frac{1}{2}, & \text{if } x \leq 0 \\ \\ 1, & \text{if } x \leq 1 \end{cases}.$$

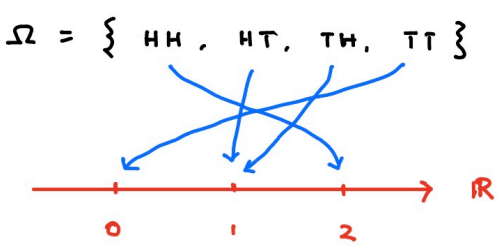


3. Expectation
$$E(X) = \sum_n x_n P_n = x_0 P_0 + x_1 P_1 = 0 + \frac{1}{2} = \frac{1}{2}.$$

4. Variance
$$\begin{aligned} Var(X) &= \sum_n (x_n - E(X))^2 P_n \\ &= (x_0 - E(X))^2 P_0 + (x_1 - E(X))^2 P_1 \\ &= \frac{1}{4}. \end{aligned}$$

5. Standard deviation
$$\sigma_X = \sqrt{Var(X)} = \frac{1}{2}.$$

Example 1.7 Consider an experiment of tossing a coin twice and count how many heads occur. Define a discrete random variable $X : \Omega \to \mathbb{R}$ where $\Omega = \{HH, HT, TH, TT\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$ such that
$$X(\omega) = \begin{cases} x_0 = 0, & \text{if } \omega = TT \\ x_1 = 1, & \text{if } \omega = HT \text{ or } TH \\ x_2 = 2, & \text{if } \omega = HH \end{cases}.$$



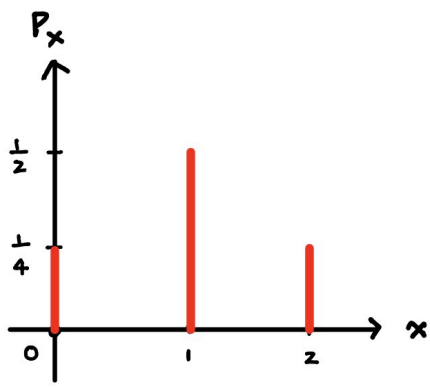
Also, define a probability measure
$$\mathbb{P}(\{HH\}) = \mathbb{P}(\{HT\}) = \mathbb{P}(\{TH\}) = \mathbb{P}(\{TT\}) = \frac{1}{4}.$$
Then we have

1. PMF

$$P_0 = P_X(x_0) = \mathbb{P}(X = 0) = \mathbb{P}(\{TT\}) = \frac{1}{4},$$
$$P_1 = P_X(x_1) = \mathbb{P}(X = 1) = \mathbb{P}(\{HT\} \cup \{TH\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$
$$P_2 = P_X(x_2) = \mathbb{P}(X = 2) = \mathbb{P}(\{HH\}) = \frac{1}{4}.$$

Thus, we get

$$P_X(x) = \begin{cases} \frac{1}{4}, & \text{if } x = 0 \\ \frac{1}{2}, & \text{if } x = 1 \\ \frac{1}{4}, & \text{if } x = 2 \end{cases}.$$

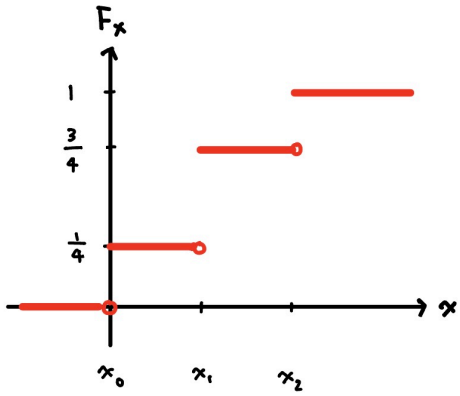


2. CDF

$$F_X(x_0) = \mathbb{P}(X \leq x_0 = 0) = P_0 = \frac{1}{4},$$
$$F_X(x_1) = \mathbb{P}(X \leq x_1 = 1) = P_0 + P_1 = \frac{3}{4},$$
$$F_X(x_2) = \mathbb{P}(X \leq x_2 = 2) = P_0 + P_1 + P_2 = 1.$$

Thus, we get

$$F_X(x) = \begin{cases} \frac{1}{2}, & \text{if } x \leq 0 \\ \frac{3}{4}, & \text{if } x \leq 1 \\ 1, & \text{if } x \leq 2 \end{cases}.$$



3. Expectation

$$E(X) = \sum_n x_n P_n = 1.$$

4. Variance

$$Var(X) = \sum_n (x_n - E(X))^2 P_n = \frac{1}{2}.$$

5. Standard deviation

$$\sigma_X = \sqrt{Var(X)} = \frac{\sqrt{2}}{2}.$$

1.6 Important Discrete Random Variables

1.6.1 Bernoulli Random Variable

The Bernoulli random variable has binary outcomes, success or failure. We assign the number 1 if an experiment has "success" occurred, and number 0 if a "failure" occurred. That is,

$$X = \begin{cases} 1, & \text{if "success"} \\ 0, & \text{if "failure"} \end{cases}.$$

Definition

Let X be a discrete random variable and its support be $R_X = [0,1]$. Then X follows a ***Bernoulli distribution*** *with parameter $p \in (0,1)$ if its probability mass function is*

$$P_X(x;p) = \begin{cases} p, & \text{if } x = 1 \\ 1-p, & \text{if } x = 0 \\ 0, & \text{if } x \notin R_X \end{cases}.$$

The cumulative distribution function of the Bernoulli R.V. is then

$$F_X(x;p) = \begin{cases} 1-p, & \text{if } x \leq 0 \\ 1, & \text{if } x \leq 1 \end{cases}.$$

Problem 1.1 Show that the expectation of a Bernoulli R.V. X is

$$E(X) = p.$$

Proof Note that

$$\begin{aligned} E(X) &= \sum_x x P_X(x) = 1 \cdot P_X(1) + 0 \cdot P_X(0) = 1 \cdot p + 0 \cdot (1-p) \\ &= p. \end{aligned}$$

Problem 1.2 Show that the variance of a Bernoulli R.V. X is

$$Var(X) = p(1-p).$$

Proof Note that

$$\begin{aligned} Var(X) &= E\left(X^2\right) - (E(X))^2 \\ &= \left(\sum_x x^2 P_X(x)\right) - \left(p^2\right) \\ &= \left(1^2 \cdot P_X(1) + 0^2 \cdot P_X(0)\right) - \left(p^2\right) \\ &= p - p^2 \\ &= p(1-p). \end{aligned}$$

1.6.2 Binomial Random Variable

The binomial random variable counts the number of "success" after n trials of a Bernoulli experiment. We have two parameters (n, p) where n is the number of repetitions of the Bernoulli experiment and p is the probability of "success" of an individual trial.

Definition

Let $Y_i, i = 1, 2, \ldots, n$ be an independent Bernoulli random variable. Define

$$S_n = \sum_{i=1}^{n} Y_n,$$

*which represents the number of "success" of Bernoulli trials. Then the discrete random variable $S_n = i$ follows a **binomial distribution** with parameters $n \in \mathbb{N}$ and $p \in (0, 1)$ if its probability mass function is*

$$P_{S_n}(i; n, p) = \begin{cases} \binom{n}{i} p^i (1-p)^{n-i}, & \text{if } i \in R_X \\ 0, & \text{if } i \notin R_X \end{cases},$$

where its support $R_X = \{0, 1, 2, \ldots, n\}$ and the binomial coefficient

$$\binom{n}{i} = \frac{n!}{i!(n-i)!}.$$

The cumulative distribution function of the binomial R.V. is then

$$F_{S_n}(i; n, p) = \mathbb{P}(S_n \leq i) = \sum_{j=0}^{i} \binom{n}{j} p^j (1-p)^{n-j}.$$

Problem 1.3 Show that the expectation of a binomial R.V. S_n is

$$E(S_n) = np.$$

Proof Note that

$$E(S_n) = \left(\sum_{i=1}^{n} Y_i \right)$$

where Y_i is an independent Bernoulli R.V.. Then

$$E(S_n) = \left(\sum_{i=1}^{n} Y_i \right) = \sum_{i=1}^{n} E(Y_i) = \sum_{i=1}^{n} p = np.$$

Problem 1.4 Show that the variance of a binomial R.V. S_n is

$$Var(S_n) = np(1-p).$$

Proof Note that

$$Var(S_n) = Var\left(\sum_{i=1}^{n} Y_i \right) = \sum_{i=1}^{n} Var(Y_i) = \sum_{i=1}^{n} p(1-p) = np(1-p).$$

3. Poisson Random Variable

The Poisson distribution counts the number of events within a specific unit, such as time, length, area, etc.. For example, the number of customers arriving at a restaurant per one hour, and the number of particles emitted by a radioactive material per second.

Let X be a discrete random variable and its support be $R_X = \mathbb{Z}_+$. Then X follows a ***Poisson distribution*** *with parameter $\lambda \in (0, \infty)$ if its probability mass function is*

$$P_X(i;\lambda) = \begin{cases} e^{-\lambda}\frac{\lambda^i}{i!}, & \text{if } i \in R_X \\ 0, & \text{if } i \notin R_X \end{cases}.$$

The cumulative distribution function of the Poisson R.V. is then

$$F_X(i;\lambda) = \mathbb{P}(X \leq i) = \sum_{j=0}^{i} e^{-\lambda}\frac{\lambda^j}{j!}.$$

Problem 1.5 Show that the expectation of a Poisson R.V. X is

$$E(X) = \lambda.$$

Proof Note first

$$E(X) = \sum_x xP_X(x) = \sum_{x=0}^{\infty} xe^{-\lambda}\frac{\lambda^x}{x!} = 0 + \sum_{x=1}^{\infty} xe^{-\lambda}\frac{\lambda^x}{x!}.$$

Let $y = x - 1$. Then

$$\begin{aligned} E(X) &= \sum_{y=0}^{\infty}(y+1)e^{-\lambda}\frac{\lambda^{y+1}}{(y+1)!} = \lambda\sum_{y=0}^{\infty} e^{-\lambda}\frac{\lambda^y}{y!} = \lambda\sum_{y=0}^{\infty} P_Y(y) \\ &= \lambda. \end{aligned}$$

Problem 1.6 Show that the variance of a Poisson R.V. X is

$$Var(X) = \lambda.$$

Proof Note first

$$\begin{aligned} Var(X) &= E\left(X^2\right) - (E(x))^2 = \sum_x x^2 P_X(x) - \left(\lambda^2\right) \\ &= \sum_{x=0}^{\infty} x^2 e^{-\lambda}\frac{\lambda^x}{x!} - \left(\lambda^2\right) \\ &= 0 + \sum_{x=1}^{\infty} x^2 e^{-\lambda}\frac{\lambda^x}{x!} - \left(\lambda^2\right). \end{aligned}$$

Let $y = x - 1$. Then

$$\begin{aligned}
Var(X) &= \sum_{y=0}^{\infty} (y+1)^2 e^{-\lambda} \frac{\lambda^{y+1}}{(y+1)!} - \left(\lambda^2\right) \\
&= \lambda \sum_{y=0}^{\infty} (y+1) e^{-\lambda} \frac{\lambda^{y}}{y!} - \left(\lambda^2\right) \\
&= \lambda \sum_{y=0}^{\infty} (y+1) P_Y(y) - \left(\lambda^2\right) \\
&= \lambda \left[\sum_{y=0}^{\infty} y P_Y(y) + \sum_{y=0}^{\infty} P_Y(y) \right] - \left(\lambda^2\right) \\
&= \lambda \left[E(Y) + 1 \right] - \left(\lambda^2\right) \\
&= \lambda^2 + \lambda - \lambda^2 \\
&= \lambda.
\end{aligned}$$